M2T1 BACHY105 · 8 Hours · Electrochemistry, Semiconductors, Li-Ion Batteries, Fuel Cells & Solar Cells

Energy Devices & Semiconductor Electrodes

Complete editorial study notes covering galvanic vs. electrolytic cells, Nernst equation, Downs cell & water electrolysis, typewise semiconductors (intrinsic, n-type, p-type, compound), zone refining, Czochralski crystal pulling, Li-ion intercalation batteries, PEMFC & SOFC fuel cells, and PV/PEC/Grätzel DSSC solar cells.

○ Mark as done ◜ Highlight Quiz this topic

All syllabus sections included · Ticks itself when every tab is read

Definitions ✓ Formulas Key Concepts Diagrams & Derivations Solved Examples

M2S0 · MEMORY STRIP

Quick-Reference Constants, Equations & Standard Potentials

Standard Cell EMF & Free Energy	**$E°_{cell} = E°_{cathode} - E°_{anode}$** — $\Delta G° = -nFE°_{cell} = -2.303\ RT \log K_{eq}$ | $F = 96{,}485\ C\ mol^{-1}$
Nernst Equation at 298.15 K (25 °C)	**$E_{cell} = E°_{cell} - (0.05916 / n) \log Q$** — Single electrode ($M^{n+} + ne^- \rightarrow M$): $E = E° + (0.05916 / n) \log [M^{n+}]$
Standard Potentials (25 °C)	**$E°_{Zn^{2+}/Zn} = -0.76$ V | $E°_{Cu^{2+}/Cu} = +0.34$ V** — $E°_{Li^+/Li} = -3.05$ V · $E°_{Fe^{2+}/Fe} = -0.44$ V · $E°_{Ag^+/Ag} = +0.80$ V
Solar Cell Fill Factor & Efficiency	**$FF = (V_{mp}I_{mp}) / (V_{oc}I_{sc})$ | $\eta = (V_{oc}I_{sc}FF) / P_{in}$** — Bandgap photon cutoff: λ_{max} (nm) = 1240 / E_g (eV) (Si: 1.1 eV → 1127 nm)

M2S1 · 8-POINT COMPARISON & 3 SALT BRIDGE FUNCTIONS

Electrochemical vs. Electrolytic Cells, IUPAC Cell Notation & Salt Bridge

Feature / Parameter	Electrochemical (Galvanic / Voltaic) Cell	Electrolytic Cell
Energy Conversion	Converts spontaneous **chemical energy → electrical energy**	Uses external **electrical energy → non-spontaneous chemical reaction**
Thermodynamics	**Spontaneous redox reaction:** $\Delta G < 0$ and $E_{cell} > 0$	**Non-spontaneous redox reaction:** $\Delta G > 0$ and $E_{cell} < 0$ (requires $V_{ext} > \|E_{cell}\|$)
Anode Sign & Process	**Negative (−) terminal** — site of **Oxidation** (*LOAN: Left, Oxidation, Anode, Negative*)	**Positive (+) terminal** — connected to + pole of DC source; site of **Oxidation**
Cathode Sign & Process	**Positive (+) terminal** — site of **Reduction**	**Negative (−) terminal** — connected to − pole of DC source; site of **Reduction**
Electrode Compartments	Two half-cells separated by a **salt bridge** or porous diaphragm	Both electrodes commonly immersed in a **single electrolyte container**
Electron Flow	Anode (−) → Cathode (+) through external load circuit	External DC battery forces electrons into Cathode (−) and pulls from Anode (+)

Feature / Parameter	Electrochemical (Galvanic / Voltaic) Cell	Electrolytic Cell
Classic Examples	Daniell cell (Zn | Zn^{2+} || Cu^{2+} | Cu, E° = 1.10 V), Dry cell, Lead-acid, Li-ion	Downs cell (molten NaCl electrolysis), Electroplating, Water splitting, Electrorefining

Daniell Cell Reactions & IUPAC Notation
E°cell = 1.10 V

Anode (Oxidation, Left): $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ (**$E°_{Zn^{2+}/Zn} = -0.76$ V**; Zn rod dissolves).

Cathode (Reduction, Right): $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ (**$E°_{Cu^{2+}/Cu} = +0.34$ V**; Cu plates onto cathode).

Overall Reaction & Notation: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ | **Zn(s) | $ZnSO_4$(1 M) || $CuSO_4$(1 M) | Cu(s)**, where **|** is a phase boundary and **||** is the salt bridge; $E°_{cell} = +0.34 - (-0.76) =$ **+1.10 V**.

Salt Bridge Composition & 3 Functions
CAT-1 / FAT 3M

Construction: Inverted U-tube filled with concentrated inert electrolyte (**KCl, KNO_3, or NH_4NO_3**) gelled in **agar-agar**. KCl/KNO_3 are chosen because **ionic mobilities (transport numbers) of K^+ and Cl^- / NO_3^- are nearly equal**. (Use KNO_3 when Ag^+, Pb^{2+}, or Hg_2^{2+} are present to avoid precipitation).

1. Completes the Internal Circuit: Allows ion migration between half-cells without mixing bulk electrolytes.

2. Maintains Electrical Neutrality: Sends anions (Cl^-/NO_3^-) into the anode compartment to balance excess Zn^{2+} and cations (K^+) into the cathode compartment to replace consumed Cu^{2+}.

3. Minimizes Liquid Junction Potential (E_j): Equal cation/anion velocities prevent asymmetric charge build-up at the liquid-liquid interface.

Effect of Opposing External Voltage (E_{ext})
Reversibility

E_{ext} < 1.10 V: Continues to act as a **Galvanic cell** (Zn dissolves, Cu deposits; e^- flow Zn → Cu).

E_{ext} = 1.10 V: Cell reaches **equilibrium**; no current flows (I = 0) and no chemical reaction occurs.

E_{ext} > 1.10 V: Polarity reverses and cell operates as an **Electrolytic cell** (Cu oxidizes at anode, Zn^{2+} reduces at cathode; e^- flow Cu → Zn).

M2S2 · DERIVATION + 2 ELECTROLYSIS SYSTEMS

Nernst Equation & Industrial Electrolysis (Molten NaCl Downs Cell & Water)

Nernst Equation Derivation
Thermodynamic Basis

Step 1 (Van't Hoff Isotherm): For a general redox reaction $aA + bB \rightleftharpoons cC + dD$ transferring n moles of e^-: **$\Delta G = \Delta G° + RT \ln Q$**, where $Q = ([C]^c[D]^d)/([A]^a[B]^b)$ (pure solids/liquids have activity = 1).

Step 2 (Electrical Work): Substitute $\Delta G = -nFE_{cell}$ and $\Delta G° = -nFE°_{cell}$: $-nFE_{cell} = -nFE°_{cell} + RT \ln Q$.

Step 3 (Divide by −nF & convert to $\log_{10}$ at T = 298.15 K): $E_{cell} = E°_{cell} - (2.303\ RT/nF) \log_{10} Q = E°_{cell} - (0.05916/n) \log_{10} Q$.

At Equilibrium: $\Delta G = 0$ and $E_{cell} = 0 \Rightarrow$ **$E°_{cell} = (0.05916/n) \log_{10} K_{eq}$** or $\log_{10} K_{eq} = (n\ E°_{cell}) / 0.05916$.

1. Electrolysis of Molten NaCl

Downs Cell (~801 °C)

Setup: Pure NaCl melts at ~801 °C (in practice mixed with $CaCl_2$ flux to lower MP to ~600 °C) using an **inert circular iron cathode** and **central graphite anode** separated by a steel gauze diaphragm.

Cathode (−, Reduction): $2Na^+(l) + 2e^- \rightarrow 2Na(l)$ ($E° = -2.71$ V; molten sodium metal collected under oil).

Anode (+, Oxidation): $2Cl^-(l) \rightarrow Cl_2(g)\uparrow + 2e^-$ ($E° = +1.36$ V; chlorine gas evolved at graphite anode).

Overall Reaction: $2NaCl(l) \rightarrow 2Na(l) + Cl_2(g)$ | **$E°_{cell} = -2.71 - 1.36 = -4.07$ V** (requires external DC voltage **> 4.07 V**). *Note: If aqueous NaCl is electrolyzed instead, H_2O reduces preferentially over Na^+ to give $H_2(g) + OH^-$ (Chlor-alkali process).*

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2. Electrolysis of Water

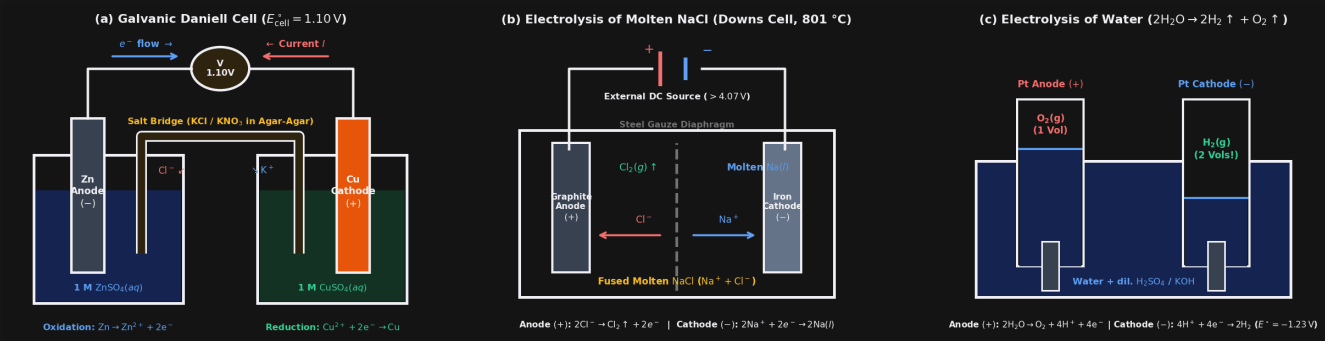
2:1 H_2 to O_2 Stoichiometry

Electrolyte: Pure water has very low conductivity ($[H^+] = 10^{-7}$ M), so dilute **H_2SO_4** (acidic) or **KOH/NaOH** (alkaline) is added with inert **Pt electrodes**.

Acidic Medium: Anode (+): $2H_2O(l) \rightarrow O_2(g) + 4H^+ + 4e^-$ ($E°_{ox} = -1.23$ V) | Cathode (−): $4H^+ + 4e^- \rightarrow 2H_2(g)$ ($E°_{red} = 0.00$ V).

Alkaline Medium: Anode (+): $4OH^- \rightarrow O_2(g) + 2H_2O + 4e^-$ | Cathode (−): $4H_2O + 4e^- \rightarrow 2H_2(g) + 4OH^-$.

Net Reaction & Voltage: $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$; **$E°_{cell} = -1.23$ V** (practical decomposition needs **1.8–2.0 V** due to oxygen overpotential on Pt). Volume of **H_2 at cathode is exactly twice (2:1)** the volume of O_2 at anode.



Visual Summary M2.1: *(Left) Galvanic Daniell cell showing Zn anode oxidation, Cu cathode reduction, and KCl salt bridge ion migration; (Center) Molten NaCl Downs cell electrolysis requiring >4.07 V; (Right) Acidulated water electrolysis yielding a 2:1 volume ratio of H_2 (cathode) to O_2 (anode).*

M2S3 · COMPLETE TYPEWISE BREAKDOWN + 3-STAGE PURIFICATION

Semiconductor Electrode Materials: Typewise Classification, Si Polymorphs & Single-Crystal Growth

1. Intrinsic Semiconductors
Pure Group IV ($n = p = n_i$)

Definition & Bonding: Chemically pure tetravalent semiconductors — **Silicon (Si, E_g = 1.12 eV)** and **Germanium (Ge, E_g = 0.67 eV)** — crystallized in a **diamond cubic tetrahedral lattice** where each atom shares 4 valence electrons via covalent bonds.

Conduction Mechanism: At 0 K, the valence band (VB) is completely full and conduction band (CB) is empty (acts as an insulator). At T > 0 K, thermal excitation breaks covalent bonds, generating equal concentrations of **electrons in CB and holes in VB ($n = p = n_i$)**. Fermi level lies exactly at the mid-gap: $E_F = (E_C + E_V)/2$.

2. Extrinsic Semiconductors
n-Type (Group V) & p-Type (Group III)

n-Type (Pentavalent Group V Donor Doping — P, As, Sb): Four valence electrons of P form covalent bonds with 4 neighbouring Si atoms; the **5th extra electron is loosely bound ($E_d \approx 0.045$ eV below E_C in Si)** and easily ionizes into the CB, leaving behind an immobile positively charged **P^+ donor ion**. Majority carriers: **electrons ($n \gg p$)**; Fermi level shifts **upward near E_C**.

p-Type (Trivalent Group III Acceptor Doping — B, Al, Ga, In): Three valence electrons of B form 3 covalent bonds with Si, leaving the **4th Si–B bond incomplete with a positive Hole (h^+)**. An acceptor level forms just above the VB ($E_a \approx 0.045$ eV **above E_V**); neighbouring valence electrons jump in, creating mobile **holes in VB** and immobile **B^- acceptor ions**. Majority carriers: **holes ($p \gg n$)**; Fermi level shifts **downward near E_V**.

Mass-Action Law (Both Intrinsic & Extrinsic): At thermal equilibrium, $n \cdot p = n_i^2$.

3. Compound Semiconductors
III–V & II–VI Direct Bandgap

III–V Compounds: Formed by combining Group III (B, Al, Ga, In) and Group V (N, P, As, Sb) elements — e.g., **GaAs (E_g = 1.42 eV, direct bandgap)**, **InP (1.35 eV)**, **GaN (3.4 eV)**, **InSb**. High electron mobility and direct radiative recombination make them ideal for high-efficiency solar cells, LEDs, laser diodes, and microwave ICs.

II–VI & Oxide Photoelectrodes: CdS (2.42 eV), **CdTe (1.45 eV)**, **ZnO (3.37 eV)**, and **TiO_2 (3.2 eV, n-type photoanode)** used in thin-film PV, PEC water splitting, and Grätzel DSSCs.

4. Structural Polymorphs of Silicon
Single vs. Poly vs. Amorphous

Monocrystalline (Single-Crystal) Si (c-Si): Continuous 3D diamond lattice with **zero grain boundaries**; highest carrier mobility and highest commercial solar cell efficiency (**20–24%**), but energy-intensive (Czochralski growth).

Polycrystalline (Multicrystalline) Si (mc-Si): Composed of multiple small crystal grains separated by **grain boundaries** that act as recombination traps; lower cost, block-cast, efficiency **15–18%**.

Amorphous Silicon (a-Si:H): Non-crystalline disordered network passivated with 5–10% hydrogen; **quasi-direct bandgap (~1.7 eV)** with 40× higher visible light absorption than c-Si, allowing 1 μm thin flexible films on glass/plastic (efficiency **7–10%**; used in calculators and thin-film panels).

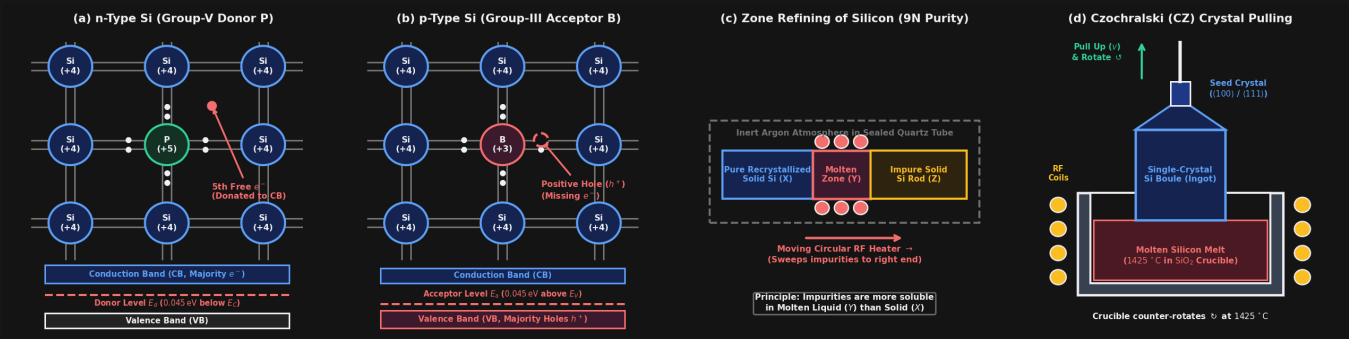
5. Silicon Purification & Single-Crystal Growth

Distillation → Zone Refining → Czochralski

Stage 1 — Carbothermic Reduction & Fractional Distillation (Siemens Process): Quartzite sand (SiO_2) is reduced with coke in an electric arc furnace ($SiO_2 + 2C \rightarrow Si + 2CO$ at 1900 °C) to give ~98% metallurgical-grade Si (MGS). MGS is converted into volatile liquid **trichlorosilane ($SiHCl_3$, BP = 31.8 °C)** via **$Si + 3HCl(g) \rightarrow SiHCl_3 + H_2$**, purified by multi-stage fractional distillation, and reduced with pure H_2 at 1100 °C (**$SiHCl_3 + H_2 \rightarrow Si + 3HCl$**) to obtain electronic-grade polysilicon.

Stage 2 — Zone Refining (99.9999999% / 9N Purity): Based on the principle that **impurities are more soluble in the molten liquid phase than in the solid crystal phase (segregation coefficient $k = C_{solid}/C_{liquid} < 1$)**. A circular RF induction heater moves slowly along a polysilicon rod under inert Ar atmosphere; the molten zone sweeps impurities to one end, which is sawed off.

Stage 3 — Czochralski (CZ) Single-Crystal Pulling & Wafer Dicing: Ultra-pure Si is melted at **1425 °C** in a quartz crucible inside an inert Ar chamber. An oriented single-crystal **seed crystal** (e.g., ⟨100⟩ or ⟨111⟩) is dipped into the melt and slowly **pulled upward while rotating** in the opposite direction to the crucible, growing a cylindrical single-crystal ingot (*boule*). The boule is sliced with diamond wire saws, lapped, and polished into mirror-smooth wafers.



Visual Summary M2.2 (Exact Slide Diagrams): *(Top Left) 2D Si covalent lattice doped with Group-V Phosphorus showing the 5th free electron and donor level E_d below E_C; (Top Right) 2D Si lattice doped with Group-III Boron showing the incomplete covalent bond (positive hole h^+) and acceptor level E_a above E_V; (Bottom Left) Single-crystal vs. polycrystalline vs. amorphous Si; (Bottom Center) Zone refining RF induction heater sweeping impurities into the melt; (Bottom Right) Czochralski single-crystal pulling from a 1425 °C Si melt.*

M2S4 · INTERCALATION CHEMISTRY, 6 VARIANTS & SAFETY

Battery Classification & Lithium-Ion Secondary Batteries (LIB)

Why Lithium? & Core Architecture
3.7 V Nominal Voltage

3 Reasons Lithium is Ideal: (1) Lightest metal in the periodic table (**M = 6.94 g mol^{-1}, ρ = 0.534 g cm^{-3}**); (2) Most negative standard reduction potential (**$E°_{Li^+/Li}$ = −3.05 V vs. SHE**), yielding **3.6–3.7 V per cell** (equivalent to three 1.2 V Ni-Cd/Ni-MH cells); (3) Tiny ionic radius of **Li^+ (0.76 Å)** allows reversible topotactic **intercalation/de-intercalation** between layered host lattices without structural collapse.

Anode (− on discharge): Layered **Graphite (C_6)** coated on a **Copper (Cu) foil** current collector.

Cathode (+ on discharge): Layered transition-metal oxide **$LiCoO_2$** (or $LiFePO_4$ / NMC) coated on an **Aluminum (Al) foil** current collector.

Non-Aqueous Electrolyte: 1 M $LiPF_6$ (or $LiBF_4$ / $LiClO_4$) dissolved in an anhydrous organic carbonate mixture — **Ethylene Carbonate (EC) + Dimethyl Carbonate (DMC) / Diethyl Carbonate (DEC)**. (Water cannot be used because Li reacts violently with H_2O and 3.7 V exceeds water's 1.23 V stability window).

Separator & SEI Layer: Microporous polyolefin (**PE/PP**, 20–25 μm) membrane isolates electrodes while passing Li^+. During the first charge, a passivating **Solid Electrolyte Interphase (SEI)** forms on the graphite anode, preventing continuous electrolyte decomposition.

Discharging & Charging Reactions
"Rocking-Chair" Mechanism

During Discharging (Spontaneous, Galvanc Mode — Li^+ rocks from Anode → Cathode):

Anode (−, Oxidation / De-intercalation): Li_xC_6 → 6C (graphite) + xLi^+ + xe^-

Cathode (+, Reduction / Intercalation): $Li_{1-x}CoO_2$ + xLi^+ + xe^- → $LiCoO_2$ (Co^{4+} reduces to Co^{3+})

Net Discharge Reaction: Li_xC_6 + $Li_{1-x}CoO_2$ → $LiCoO_2$ + 6C (E_{cell} ≈ 3.7 V)

During Charging (Non-Spontaneous, Electrolytic Mode — External DC forces Li^+ from Cathode → Anode):

Positive Electrode (Anode of charging, Oxidation): $LiCoO_2$ → $Li_{1-x}CoO_2$ + xLi^+ + xe^-

Negative Electrode (Cathode of charging, Reduction): 6C + xLi^+ + xe^- → Li_xC_6

7 Key Advantages & 3 Thermal Runaway Causes
Exam Short-Note

7 Advantages: (1) High cell voltage (**3.7 V**); (2) High gravimetric energy density (**150–260 Wh/kg**); (3) **Zero memory effect** (unlike Ni-Cd); (4) Low self-discharge (**< 2–5% per month**); (5) High Coulombic efficiency (>99%); (6) Long cycle life (1000–3000 cycles); (7) Light weight & no toxic Cd/Pb/Hg.

3 Causes of LIB Fires / Explosions (Thermal Runaway): (1) **Lithium Dendrite Growth:** Overcharging or fast charging at low T plates metallic Li whiskers on the anode that pierce the PE/PP separator, causing an internal short circuit; (2) **Cathode Oxygen Release:** Above 180 °C, delithiated $Li_{1-x}CoO_2$ decomposes exothermically, releasing O_2(g) inside the sealed cell; (3) **Flammable Organic Carbonate Electrolyte:** Low-flash-point DMC/DEC/EC vapors ignite upon contact with released O_2 and heat, rupturing the casing.

LIB Cathode / Anode Variant	Formula & Structure	Nominal Voltage & Energy Density	Key Features & Primary Applications
1. LCO (Lithium Cobalt Oxide)	$LiCoO_2$ (Layered oxide)	3.7 V · 150–200 Wh/kg	High specific energy, lower thermal stability; smartphones, laptops, cameras
2. LMO (Lithium Manganese Oxide)	$LiMn_2O_4$ (3D Spinel)	3.7 V · 100–150 Wh/kg	Low internal resistance, high discharge rate; power tools, medical devices
3. LFP (Lithium Iron Phosphate)	$LiFePO_4$ (Olivine)	3.2–3.3 V · 90–160 Wh/kg	Strong P–O covalent bond prevents O_2 release; safest, 3000+ cycles; EVs & grid storage
4. NMC (Nickel Manganese Cobalt)	$LiNi_xMn_yCo_zO_2$	3.6–3.7 V · 150–240 Wh/kg	Ni gives capacity, Mn gives spinel stability, Co gives rate; dominant EV battery
5. NCA (Nickel Cobalt Aluminum)	$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$	3.6 V · 200–260 Wh/kg	Highest energy density among commercial cathodes; long-range EVs
6. LTO (Lithium Titanate Anode)	$Li_4Ti_5O_{12}$ (replaces graphite)	2.4 V · 50–80 Wh/kg	Zero-strain anode, no SEI/dendrites, ultra-fast 10-min charge, 10,000+ cycles

M2S5 · 8 FUEL CELL TYPES + DEEP PEMFC & SOFC

Fuel Cells: Primary vs. Secondary vs. Fuel Cells, PEMFC & SOFC

Feature	Primary Battery	Secondary (Storage) Battery	Fuel Cell
Reversibility	Irreversible cell reaction; discarded after single discharge	Reversible cell reaction; recharged hundreds/thousands of times by external DC	Reactants continuously supplied externally; never needs electrical recharging
Reactant Storage	Fixed mass of active species sealed inside cell	Fixed mass of active species sealed inside cell (energy storage device)	Fuel (H_2, CO, CH_4) and oxidant (O_2/air) stored in external tanks (energy conversion device)
Examples	Leclanché dry cell (Zn–MnO_2), Alkaline cell, Zn–Ag_2O button cell	Lead-acid, Ni–Cd, Ni–MH, Lithium-ion (LIB)	PEMFC (Nafion, <80 °C), SOFC (YSZ, 800–1000 °C), AFC, PAFC, MCFC, DMFC

Overview of 8 Fuel Cell Families
Electrolyte & Temp

Low-Temperature (< 100 °C): (1) **PEMFC** (Nafion sulfonated PTFE membrane, H^+ ion, 60–80 °C); (2) **AFC** (Aqueous KOH, OH^- ion, 60–90 °C; Apollo spacecraft); (3) **DMFC** (Direct Methanol $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$, Nafion, 50–100 °C); (4) **DEFC** (Direct Ethanol) & **DFAFC** (Direct Formic Acid).

Medium/High-Temperature (150–1000 °C): (5) **PAFC** (Liquid H_3PO_4 in SiC matrix, H^+, 180–210 °C); (6) **MCFC** (Molten Li_2CO_3/K_2CO_3 in $LiAlO_2$, CO_3^{2-} ion, 600–650 °C); (7) **SOFC** (Dense YSZ ceramic, O^{2-} ion, 800–1000 °C).

1. Proton Exchange Membrane Fuel Cell (PEMFC)
Low-T (< 80 °C) · Mobile H^+

Electrolyte Membrane: Solid fluoropolymer cation-exchange membrane — **Nafion** (sulfonated tetrafluoroethylene copolymer with hydrophilic **$-SO_3^-H^+$** clusters that conduct **H^+ protons** when hydrated while blocking electrons and gases).

Electrodes & Catalyst: Porous carbon cloth/paper gas-diffusion electrodes (GDL) loaded with **Pt nanoparticles (or Pt–Ru alloy)** at the triple-phase boundary.

Anode (–, Oxidation): $2H_2(g) \rightarrow 4H^+ + 4e^-$ (H^+ migrates through hydrated Nafion from Anode → Cathode).

Cathode (+, Reduction): $O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O(l)$ | **Net:** $2H_2 + O_2 \rightarrow 2H_2O$ ($E° = 1.23$ V; practical **0.7–0.8 V** per cell).

Pros & Cons: Rapid cold start-up, solid non-corrosive electrolyte, zero emissions (pure water byproduct used as **astronaut drinking water in Gemini/space missions** & FCEVs like Toyota Mirai). *Limitations:* Expensive Pt catalyst is **poisoned by >10 ppm CO**; requires strict water management (operating **< 80 °C** so Nafion does not dehydrate and lose proton conductivity).

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2. Solid Oxide Fuel Cell (SOFC)
High-T (800–1000 °C) · Mobile O^{2-}

Solid Ceramic Electrolyte: Dense gas-tight **YSZ (Yttria-Stabilized Zirconia: ZrO_2 doped with 8 mol% Y_2O_3)**. Doping trivalent Y^{3+} into tetravalent Zr^{4+} sites creates **oxygen ion vacancies (V_O)**, enabling rapid conduction of **oxide ions (O^{2-}) from Cathode → Anode at 800–1000 °C**.

Anode (–): Porous **Ni–YSZ cermet** (nickel-zirconia composite) — no precious Pt needed!

Cathode (+): Porous perovskite **LSM (Lanthanum Strontium Manganite, $La_{1-x}Sr_xMnO_3$)**.

Cathode (+, Reduction): $O_2(g) + 4e^- \rightarrow 2O^{2-}$ (O^{2-} ions hop through YSZ vacancies from Cathode → Anode).

Anode (–, Dual-Fuel Syngas Oxidation): Oxidizes both H_2 and CO (CO is a fuel, NOT a poison!):

$H_2(g) + O^{2-} \rightarrow H_2O(g) + 2e^-$ and **$CO(g) + O^{2-} \rightarrow CO_2(g) + 2e^-$**

Overall Syngas Reaction: $H_2(g) + CO(g) + O_2(g) \rightarrow H_2O(g) + CO_2(g)$

Pros & Cons: Fuel-flexible (internal reforming of CH_4/biogas/syngas), non-precious Ni catalyst, no liquid flooding, **60% electrical efficiency (85% with CHP waste-heat co-generation)**. *Limitations:* Slow thermal start-up, thermal expansion mismatch, and ceramic sealing challenges at 1000 °C.

M2S6 · 3 SOLAR ARCHITECTURES + 10-POINT COMPARISON

Solar Energy Conversion: Silicon p–n PV Cells, PEC Cells & Grätzel DSSC

1. Silicon p–n Junction Solar Cell (PV)
Solid-State Depletion Field

Construction: Antireflection-coated (Si_3N_4/TiO_2) glass cover, front metallic finger grid, thin heavily doped **n-type Si emitter (~0.3 µm)**, thicker **p-type Si base (~200 µm)**, and continuous Al back contact.

Built-in Electric Field: At the p–n interface, electrons diffuse from n → p and holes from p → n, leaving behind uncovered **positively charged donor ions (P^+) on the n-side** and **negatively charged acceptor ions (B^-) on the p-side**. This forms a carrier-free **depletion region (space-charge layer)** with a built-in electric field $E_{built\text{-}in}$ pointing from **n → p**.

Photovoltaic Action: Incident photons with **$h\nu \geq E_g$ (1.12 eV for Si, $\lambda \leq 1100$ nm)** generate electron–hole pairs. $E_{built\text{-}in}$ sweeps **electrons to the n-type front contact (negative terminal)** and **holes to the p-type back contact (positive terminal)**, driving DC current through an external load ($V_{oc} \approx 0.6$ V, $\eta \approx 18$–24%).

2. Photoelectrochemical (PEC) Cells
Semiconductor–Electrolyte Interface

Principle: Replaces the solid-state p–n junction with a **semiconductor–liquid electrolyte interface**. Equilibration between the semiconductor Fermi level (E_F) and redox couple potential (E_{redox}) creates upward **band bending** (for n-type photoanode) and a surface space-charge layer that separates photogenerated e^-/h^+ pairs without needing ultra-high-purity single crystals.

Two Sub-Classes: (a) **Liquid-Junction Solar Cells (LJSC / Regenerative PEC):** Converts light → electricity with **$\Delta G = 0$** via a reversible redox shuttle (e.g., I^-/I_3^- or S^{2-}/S_n^{2-}); (b) **Photoelectrosynthetic (PES) Cells:** Drives non-spontaneous fuel-forming reactions (**$\Delta G > 0$**) such as **Honda–Fujishima photoelectrolysis of water on n-TiO_2** ($2H_2O + 4h\nu \rightarrow 2H_2 + O_2$) or CO_2 reduction.

4 Mandatory Semiconductor Photoelectrode Conditions: (1) Optimal bandgap **$E_g = 1.4$–2.4 eV** to harvest visible light; (2) Resistance to **photocorrosion** in aqueous electrolyte; (3) Band edges must **straddle the redox potentials** (E_{CB} more negative than H^+/H_2 at 0.0 V; E_{VB} more positive than O_2/H_2O at +1.23 V); (4) Fast interfacial charge transfer and low cost.

3. Dye-Sensitized Solar Cell (DSSC / Grätzel Cell)

5 Components & 4-Step Cycle

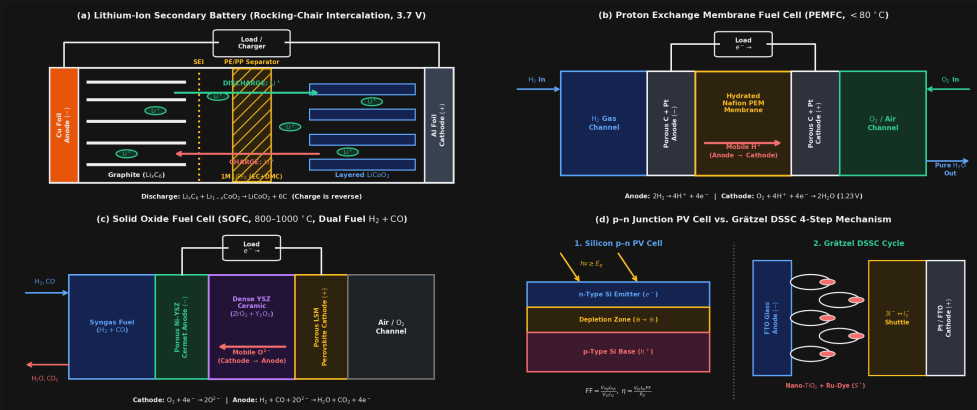
Key Innovation: Decouples **light absorption** (performed by a molecular ruthenium/organic dye monolayer S) from **charge transport** (electron transport in mesoporous n-TiO_2; hole/ion transport in I^-/I_3^- electrolyte).

5 Core Components: (1) Transparent conducting oxide anode (**FTO or ITO glass**); (2) Mesoporous wide-bandgap **nanocrystalline TiO_2 film (E_g = 3.2 eV, 10–20 nm particles)** giving ~1000× surface area; (3) Monolayer of **Ru-polypyridyl sensitizer dye (N3 / N719)** covalently anchored via −COOH groups; (4) **Iodide/Triiodide (I^-/I_3^-)** organic liquid electrolyte; (5) **Pt-coated FTO counter electrode (cathode)**.

4-Step Operating Mechanism:

1. **Photoexcitation: S + hν → S*** (visible photon promotes dye electron from HOMO → LUMO).
2. **Ultrafast Electron Injection (fs–ps): $S^* \rightarrow S^+ + e^-_{(CB\ of\ TiO_2)}$** (electron percolates through TiO_2 → FTO anode → external load → Pt cathode).
3. **Dye Regeneration by Iodide: $2S^+ + 3I^- \rightarrow 2S + I_3^-$** (prevents $S^+ + e^-$ back-recombination).
4. **Electrolyte Regeneration at Pt Cathode: $I_3^- + 2e^-_{(Pt)} \rightarrow 3I^-$** (completes closed regenerative cycle).

Parameter	Conventional Silicon p–n PV Cell	Photoelectrochemical (PEC) Cell	Dye-Sensitized Solar Cell (DSSC)
Light Absorber	Bulk Si semiconductor (E_g = 1.12 eV)	Semiconductor electrode (TiO_2, CdSe, Fe_2O_3)	Molecular dye monolayer (Ru-N719) on TiO_2
Charge Separation	Built-in field at solid p–n homojunction	Band bending at semiconductor–liquid interface	Ultrafast LUMO → TiO_2 CB electron injection
Purity Requirement	Ultra-high 9N purity single/poly-Si required	Moderate; polycrystalline films work well	Low-purity, low-cost nanocrystalline TiO_2
Diffuse / Indoor Light	Efficiency drops sharply at low/angled light	Moderate performance	Excellent performance in diffuse, cloudy & indoor light
Output Product	DC Electricity only	DC Electricity (LJSC) OR Chemical fuel H_2 (PES)	DC Electricity



Visual Summary M2.3 (Slide Diagrams): *(Top Left) Li-ion battery rocking-chair Li^+ intercalation between Graphite (C_6) and $LiCoO_2$; (Top Center) PEMFC showing H^+ transport across hydrated Nafion (<80 °C); (Top Right) SOFC showing O^{2-} vacancy transport across YSZ ceramic (800–1000 °C) with dual H_2 + CO fuel; (Bottom Left) Silicon p–n junction PV cell; (Bottom Right) Grätzel DSSC energy-level diagram and 4-step I^-/I_3^- regenerative cycle.*

M2S7 · 6 NUMERICALS + 14 PYQ SOLUTIONS

Complete Solved Numericals & Official VIT Question Bank Blueprints (Q1–Q14)

Solved Numerical 2.1 — Nernst Equation, Cell EMF & ΔG for Fe | Fe^{2+} || Cu^{2+} | Cu
Official Slide Numerical · 5 Marks

Problem: Calculate the EMF and free energy change (ΔG) at 25 °C for the cell: **Fe(s) | Fe^{2+}(0.1 M) || Cu^{2+}(0.01 M) | Cu(s)**. Given: $E°_{Fe^{2+}/Fe}$ = −0.44 V and $E°_{Cu^{2+}/Cu}$ = +0.34 V.

1. Standard EMF ($E°_{cell}$): Anode (Oxidation): Fe → Fe^{2+} + $2e^-$; Cathode (Reduction): Cu^{2+} + $2e^-$ → Cu (n = 2).

$E°_{cell} = E°_{cathode} - E°_{anode} = +0.34 - (-0.44) = +0.78$ V.

2. Cell EMF via Nernst Equation: Q = $[Fe^{2+}]/[Cu^{2+}]$ = 0.1 / 0.01 = 10.

$E_{cell} = 0.78 - (0.05916 / 2) \log_{10}(10) = 0.78 - 0.02958(1) = +0.7504$ V.

3. Gibbs Free Energy (ΔG): $\Delta G = -nFE_{cell}$ = −2 × 96,485 C mol^{-1} × 0.7504 V = **−144,804 J mol^{-1} = −144.80 kJ mol^{-1}**.

Solved Numerical 2.2 — Asymmetric Stoichiometry (n = 2): Zn | Zn^{2+}(0.2 M) || Ag^+(0.002 M) | Ag
CAT-1 / FAT PYQ · 5 Marks

Problem: Calculate $E°_{cell}$, E_{cell} at 298 K, and equilibrium constant K_{eq} for **Zn(s) | Zn^{2+}(0.2 M) || Ag^+(0.002 M) | Ag(s)** given $E°_{Zn^{2+}/Zn}$ = −0.76 V and $E°_{Ag^+/Ag}$ = +0.80 V.

1. Balanced Reaction (n = 2): Zn(s) + $2Ag^+$(aq) → Zn^{2+}(aq) + 2Ag(s). Note that **$[Ag^+]$ is squared** in Q!

2. Standard EMF: $E°_{cell} = +0.80 - (-0.76) = +1.56$ V.

3. Nernst Equation: Q = $[Zn^{2+}] / [Ag^+]^2$ = 0.2 / $(0.002)^2$ = 0.2 / (4×10^{-6}) = 5×10^4 ⇒ $\log_{10}$ Q = 4.6990.

$E_{cell} = 1.56 - (0.05916 / 2) \times 4.6990 = 1.56 - 0.1390 = +1.421$ V.

4. Equilibrium Constant: $\log_{10} K_{eq} = (n E°_{cell})/0.05916$ = (2 × 1.56)/0.05916 = 52.74 ⇒ **$K_{eq} = 5.5 \times 10^{52}$**.

Solved Numerical 2.3 — Finding Unknown Ion Concentration & Concentration Cell EMF
Official Question Bank Q3 · 5 Marks

Problem: (a) A Daniell cell **Zn | $ZnSO_4$(x M) || $CuSO_4$(0.02 M) | Cu** registers E_{cell} = 1.07 V at 298 K ($E°_{cell}$ = 1.10 V). Find *x*. (b) Find the EMF of the concentration cell **Cu | Cu^{2+}(0.001 M) || Cu^{2+}(0.1 M) | Cu** at 298 K.

(a) Unknown $[Zn^{2+}]$ = x: 1.07 = 1.10 − (0.05916 / 2) $\log_{10}$(x / 0.02) ⇒ $\log_{10}$(x / 0.02) = 0.03 / 0.02958 = 1.0142 ⇒ x / 0.02 = $10^{1.0142}$ = 10.33 ⇒ **x = $[Zn^{2+}]$ = 0.207 M**.

(b) Concentration Cell ($E°_{cell}$ = 0 V): E_{cell} = (0.05916 / 2) $\log_{10}(C_2 / C_1)$ = 0.02958 $\log_{10}$(0.1 / 0.001) = 0.02958 × 2 = **+0.0592 V**.

Solved Numerical 2.4 — Solar Cell Fill Factor, Efficiency & Li-Ion Specific Energy
FAT Numerical · 5 Marks

Problem: (a) A 4 cm^2 silicon solar cell under 100 mW cm^{-2} illumination gives V_{oc} = 0.62 V, I_{sc} = 140 mA, V_{mp} = 0.50 V, and I_{mp} = 128 mA. Calculate Fill Factor (FF) and power conversion efficiency (η). (b) A 3.7 V $LiCoO_2$ pouch cell delivers 4.0 Ah and weighs 74 g. Calculate stored energy (Wh) and specific energy (Wh kg^{-1}).

(a) Solar Cell: $P_{max} = V_{mp}I_{mp}$ = 0.50 × 128 = 64.0 mW. **FF = 64.0 / (0.62 × 140) = 64.0 / 86.8 = 0.737 (73.7%)**.

Incident power P_{in} = 100 mW cm^{-2} × 4 cm^2 = 400 mW ⇒ **η = (64.0 / 400) × 100 = 16.0%**.

(b) Li-Ion Cell: Energy E = V × Q = 3.7 V × 4.0 Ah = **14.8 Wh**. Specific energy = 14.8 Wh / 0.074 kg = **200 Wh kg^{-1}**.

Official Question Bank Q1–Q14 Blueprints
Rapid Exam Recall

Q1–Q3 (Galvanic vs. Electrolytic, Salt Bridge & Nernst): Use 8-point Table M2S1, KCl/agar-agar 3 functions (circuit completion, electroneutrality, minimizing E_j), and Nernst derivation + Numericals 2.1–2.3.

Q4–Q5 (Molten NaCl & Water Electrolysis): Downs cell at 801 °C (2NaCl → 2Na + Cl_2, E° = −4.07 V) vs. Acidic/Alkaline H_2O splitting ($2H_2O$ → $2H_2$ + O_2, E° = −1.23 V, 2:1 H_2:O_2 volume ratio).

Q6–Q8 (Semiconductors, Zone Refining & Czochralski): Intrinsic (Si/Ge) vs. Extrinsic n-type (P donor, E_d) & p-type (B acceptor, E_a) + c-Si/mc-Si/a-Si + Siemens trichlorosilane distillation → Zone refining (k < 1) → Czochralski 1425 °C seed pulling (Diagram M2.2).

Q9–Q11 (Li-Ion Battery Construction, Intercalation & Safety): Graphite/Cu anode, $LiCoO_2$/Al cathode, 1 M $LiPF_6$ in EC+DMC, PE/PP separator, SEI layer, rocking-chair equations, 6 variants table, and 3 explosion causes (dendrites, CoO_2 O_2 evolution, flammable carbonate vapors).

Q12–Q14 (PEMFC vs. SOFC & PV vs. PEC vs. DSSC): PEMFC (Nafion, H^+, <80 °C, Pt catalyst, CO poison) vs. SOFC (YSZ ceramic, O^{2-} vacancies, 800–1000 °C, Ni–YSZ anode, H_2+CO fuel) + Silicon p–n PV vs. PEC (LJSC/PES) vs. Grätzel DSSC 4-step I^-/I_3^- cycle (Diagram M2.3).
